

Christos Hadjinikolis

Engineering Manager | AI Connectivity & Platform Reliability

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

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about me

profiles

[website](#)

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leadership

6 direct reports
career/performance
hiring/coaching
delivery ownership

connectivity

tool routing
approval gates
eval/replay
auditable LLMs
tool connectors

platform

distributed systems
Kafka/Flink
model/API serving
observability
incident response

trust

AI governance
open standards
auditability
explainability
enterprise trust

open source

PyPI: dynamicio
skeleton-replay
JetBrains plugin

Engineering Manager with 16 years across backend platforms, production ML, data systems, and AI research. Manages **6** direct reports and leads cross-functional teams responsible for reliable, observable model-backed services. Recent work spans streaming platforms, model and API serving, eval/replay loops, fallback paths, incident response, and AI governance; Promet adds private hands-on GenAI work on tool routing, approval gates, traces, and auditable LLM workflows.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own engineering strategy and delivery for a live post-ingestion ML/data estate turning **6M** vessel-position records/hour into production intelligence for **13.5K** monitored vessels. Manage **6** direct reports and lead a **10-person** cross-functional team accountable for model/API serving, data quality, stakeholder alignment, reliability, and delivery maturity.

- **People & Delivery:** Manage **6** direct reports across Product, SME analysis, Data Science, and Data Engineering; own 1:1s, performance/career development, hiring for **6** roles, coaching, delivery accountability, and team cadence.
- **Platform & Reliability:** Lead a distributed **Kafka/Flink** streaming estate; maintain rollback/fallback paths, shared on-call, incident response, observability/runbooks, and **MTTR** under **30** min.
- **Eval/Replay & Serving:** Led 0-to-1 research-to-production delivery using **PyTorch** for destination/arrival-time sequence and transformer models; established MLflow, model/data versioning, model/API serving, evaluation gates, replay, and monitoring.
- **Trust, Interfaces & Governance:** Drove data-lineage, schema-management, Avro/interface-contract, SLA, and domain-owner proposals so cross-team dependencies were explicit and governable.
- **Product / Stakeholder Alignment:** Ran Product/SME/analyst/engineer workshops to turn model-quality disputes into shared definitions, measurable objectives, interface proposals, and OKR-linked work-streams.
- **Engineering Ecosystem:** Standardised I/O seams, repo archetypes, AWS CodeArtifact/shared-library publishing, ADRs, dev containers, and local E2E tests (cut a **3h** DAG feedback loop to under **5** min).

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and helped grow it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped enterprise client opportunities, timelines, expectations, consultant placement, and technical roadmaps across UBS, Vodafone, CNHi, and other clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform (based on KubeFlow) delivery; drove requirements, migration planning, CI/CD, automation, and ML lifecycle support.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance (Azure/Databricks); translated stakeholder needs into DS/DE backlog and guided a **PySpark** team.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka/Elasticsearch** pipelines.

KCL, UCL, David Game College, UK 2010–2016

Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques (e.g., MCMC). Received Best Poster Award at IJCAI 2013 (413 submissions), EPSRC PhD Scholarship , Outstanding TA Award , and Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. Majored in Artificial Intelligence .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

selected AI/connectivity tooling

Private	Promet Private applied GenAI project for memory-aware, voice-enabled, tool-using assistants; hands-on work on local runtimes, MCP-style connector patterns, tool routing, schema validation, approval gates, durable state/memory, traces, streaming UX, and deterministic replay/eval for auditable LLM interactions.	GenAI harness
2026	MCP: Plumbing, Trust Practical article on MCP mental models, connector plumbing, staged tool exposure, host-side routing, approval, and evidence boundaries for local/private AI assistants.	Public writing
PyPI	skeleton-replay Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and LLM-assisted code understanding.	Runtime evidence
JetBrains	Skeleton Replay plugin Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.	IDE workflow
PyPI	dynamicio Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.	Data contracts

standards/committees

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

credentials & awards

2017–2020	Selected ML/cloud credentials AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.	AWS, Google Cloud, O'Reilly
2016–2019	Data/graph engineering credentials Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.	Coursera, Elastic, Neo4j